The Training of the SST-QCRC Model

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1. The formulation of the model

The model is based on Menter's Shear-Stress-Transport (SST) model [1]. Two modifications are made to the SST model. The first one is that a nonlinear quadratic correction term is added to the Reynolds stress's expression. The correction term (for incompressible flow) is like the QCR term used in the SA-QCR2000 [2] model:

$$\begin{aligned} &\tau_{ij} = \tau_{ij}^{l} - c_r\left(O_{ik}\tau_{kj}^{l} - \tau_{ik}^{l}O_{kj}\right) \\ &\tau_{ij}^{l} = 2\nu_T S_{ij} - \frac{2}{3}k\delta_{ij}, O_{ij} = \frac{\partial_j U_i - \partial_i U_j}{\sqrt{\partial_n U_m \partial_n U_m}} \end{aligned} \tag{1}$$

For this modification, the value of c_r is directly adopted from Ref. [2], which is $c_r = 0.3$. No data-driven techniques are used to train the parameters of the correction term.

The second modification is made to the destruction term of the ω's equation. It has the following form:

$$\begin{aligned} &\frac{D\omega}{Dt} = \frac{\gamma}{\nu_T}\tau_{ij}\partial_j U_i - \beta\theta\omega^2 + \partial_j\left[(\nu + \sigma_\omega \nu_T)\partial_j\omega\right] + 2(1 - F_1)\frac{\sigma_{\omega 2}}{\omega}\partial_j k \partial_j \omega \\ &\beta = [(\beta_{CND} - 1)f_d + 1] \\ &f_d = 1 - \tanh[(8r_d)^3] \\ &r_d = \frac{\nu + \nu_T}{\kappa^2 d^2 \sqrt{\partial_n U_m \partial_n U_m}} \end{aligned} \tag{2}$$

This is identical to the formulation used in the work of Wu et al. [3]. β_{CND} is a spatially varying field that quantifies the error in the ω's equation. f_d is a shielding function that is 0 in the boundary layer and is 1 elsewhere. f_d is used to deactivate the correction term β_{CND} in the boundary layer to preserve the baseline SST model's accuracy in simple wall-attached flows. The spatial distribution of β_{CND} is determined by field inversion method proposed by Singh et al. [4].

Since the QCR term is added to the Reynolds stress tensor, and the ω equation is corrected, we name the current model formulation **the SST-QCRC model** (the last 'C' stands for 'correction').

2. The training of the model

We performed field inversion on the curved backward facing step (CBFS) case. In fact, the data obtained in Ref. [3] using the conditioned field inversion is directly used (the formulation of the ω's equation is identical to the current report in the conditioned field inversion). The code used to perform the field inversion is DAFoam, developed by He et al [5]. The optimal distribution of the β_{CND} term is shown in Figure 1. The correction term β_{CND} is mainly activated in the separated shear layer.

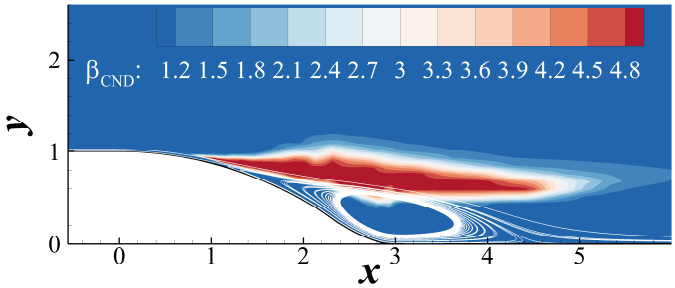

Figure 1. The optimal distribution of β_{CND} obtained by field inversion

After obtaining the data for β_{CND}, we can train a model that maps the local flow features to the value of β_{CND}. We only chose 3 features as the input features: $\lambda_1, \lambda_2, P_k/\varepsilon$. The definition of the features can be found in Ref. [3]. We use the symbolic regression algorithm to train the model. That is to say, the final model would be an explicit expression for β_{CND}. This enables us to implement the model easily and keeps the model interpretable. The PySR package [6] is used for symbolic regression. After symbolic regression, the following expressions with different complexities were distilled from the dataset, as shown in Figure 2. It can be noticed that the expression with the highest complexity (16) has a nearly identical loss compared with the expression with complexity = 15. Consequently, to keep the expression as simple as possible, we use the expression with complexity = 15 as the final expression. The final expression can be written as:

$$\beta_{CND} = \max(-0.1157, 0.0058525\lambda_2)\lambda_2 \tag{3}$$

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Hall of Fame:
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Complexity  Loss        Score      Equation
2           1.356e+00   7.971e+00  y = P2E
6           1.173e+00   3.635e-02  y = (tanh(P2E) * P2E)
7           6.063e-01   6.597e-01  y = (-0.10437 * lambda2)
8           5.764e-01   5.050e-02  y = (-0.11015 / inv1(lambda2))
10          5.591e-01   1.523e-02  y = (-0.11575 / inv1(inv(inv1(lambda2))))
11          5.591e-01   -0.000e+00  y = (tanh(-0.1163) / inv1(inv(inv1(lambda2))))
12          5.415e-01   3.208e-02  y = (-0.11499 / inv1(lambda2 * tanh(P2E)))
14          5.236e-01   1.684e-02  y = (0.12395 * (inv1(inv(tanh(tanh(lambda2)))) - lambda2))
15          4.845e-01   7.746e-02  y = (max(-0.1157, lambda2 * 0.0058525) * lambda2)
16          4.845e-01   2.253e-05  y = (max(-0.11568, tanh(lambda2 * 0.0058703)) * lambda2)
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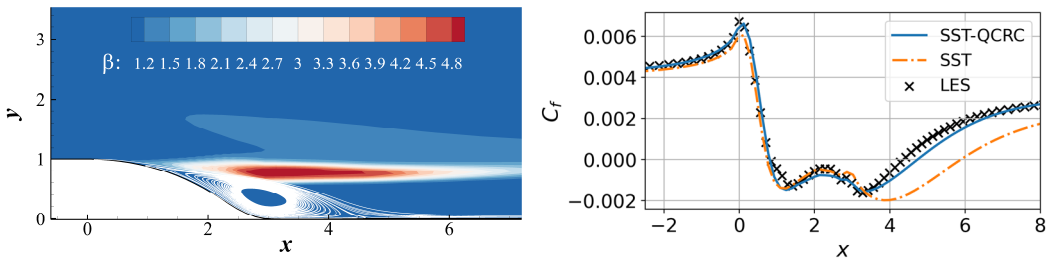
Figure 2. The result of the symbolic regression process

Finally, the expression of β_{CND} is integrated back to Eq. (2). Another trick is incorporated in this step. That is, we choose to bound the value of β_{CND} to be lower than 4 to keep the computation stable. The upper bound, that is, 4, is consistent with the upper bound used in the conditioned field inversion process. Eq. (3) implies that we choose to increase the destruction of ω in the regions with strong shear. This means that the eddy viscosity would be increased in the region of strong shear, since $\nu_T \propto k/\omega$. This is consistent with the 'slingshot effect' [7] observed in RANS simulations.

3. The performance of the model on the training set and the zero-pressure-gradient flat plate

In this section, we show the SST-QCRC model's performance in the training set (the CBFS case) and the zero-pressure-gradient (ZPG) flat plate. The ZPG flat plate flow is considered to be important by Rumsey [7] and Spalart [8] because it widely exists in engineering applications.

For the CBFS case, the correction term β and the C_f distribution on the lower surface are shown in Figure 3. The distribution of β is like the result of the conditioned field inversion, showing the effectiveness of the symbolic regression. The C_f distribution given by the SST-QCRC model fits the LES data [9] much better compared with the baseline SST model.



(a) (b)

Figure 3. The result of the SST-QCRC model in the CBFS case, (a) the distribution of the β field, (b) the C_f distribution along the lower surface

For the turbulent flat plate case, the C_f distribution is shown in Figure 4. The prediction of the SST-QCRC model is almost identical to the prediction of the SST model. Both models align well with the experiment data [10]. The result indicates that the SST-QCRC model won't affect the baseline SST model's accuracy in simple wall attached flows such as the ZPG flat plate. It also demonstrates that the shielding function f_d used in the modeling process is effective.

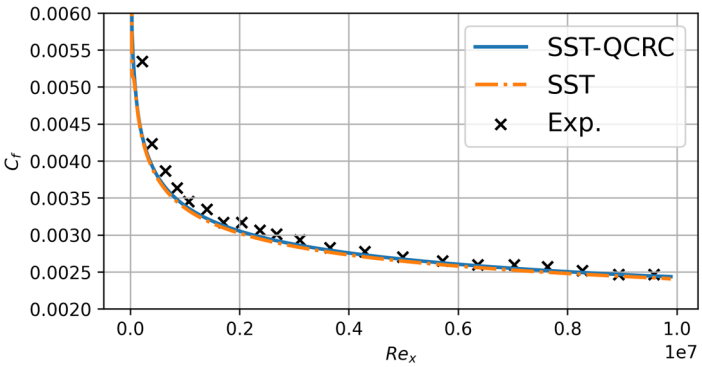


Figure 4. The C_f distribution in the turbulent flat plate case

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